

RESEARCH ARTICLE

Exhaustivity and contrastivity in Hungarian focus constructions: A visual-world study

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Abstract

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1. Introduction

How languages express contrast and exhaustivity has major implications for how we understand the syntax–semantics interface. This study revisits the debate on the interpretation of Hungarian focus constructions using eyetracking with the visual-world paradigm. Specifically, we examine two theoretically important inquiries: how strongly Hungarian native speakers interpret phrases in the pre-verbal focus domain exhaustively, and whether contrastive but non-exhaustive readings exist, as well as from a cognitive perspective, whether different focus types produce processing differences during language comprehension.

- (1) *Anna meg-ette a banánt.*
Anna[NOM] VPX-ate-3SG.DEF the banana-ACC
'Anna ate (up) the banana.'
- (2) *Anna a BANÁNT ette meg.*
Anna[NOM] the banana-ACC ate-3SG.DEF VPX
a) 'Anna ate (up) the BANANA.' [and nothing else]
b) 'Anna ate (up) the BANANA.' [and not something else]

Consider the canonical, neutral sentence in example 1, where the subject *Anna* is in the sentence-initial position, followed by the verb, *ette* 'ate' (together with the verbal prefix [VPX] (or verbal modifier [VM]) *meg-*)¹ and the object *banán* 'banana' (with the *-t* accusative ending); vs the pre-verbal focus construction in example 2, where the object is fronted to the pre-verbal position, leaving its (normally attached) verbal prefix behind. The sentence in example 2 is a typical example for a focus construction, and it could also be rendered in English with an *it*-cleft, i.e., 'It is the BANANA that Anna ate.'. The comparison of the Hungarian focus-constructions with English *it*-clefts has its own literature (cf. [É. Kiss 1999](#); [Wedgwood et al. 2006](#)), but it is out of the scope of the present discussion.

Constituents in canonical sentences can be focused for various communicative purposes, such as emphasis, question answering, negation and contradiction. In 2 we are focusing² the object, and as such we can observe two changes that mark focus: movement and prosody. Regarding the first, Hungarian has a special, grammaticalized syntactic position for the focused constituent directly in front of the verb – called the pre-verbal focus (PVF) position ([Balogh 2007](#); [Wedgwood 2006](#)) or structural focus position ([É.Kiss 1998](#)) or simply just focus position – often associated with exhaustive/identificational semantics ([Balogh 2013](#)). Besides movement, the syntactically expressed focus is also accompanied by prosodic stress, and it has been suggested that in these focus marking strategies movement is primary, and prosody is secondary ([Balogh 2013](#)). Regarding intonation in PVF constructions, the focused constituent typically bears eradicating stress (cf. [Gécseg 2020](#); [Kornai & Kálmán 1988](#)), meaning that the verb is destressed and the acoustic prominence of all other elements are 'flattened'. The PVF construction is a common, everyday focus-marking strategy in Hungarian – a topic-prominent, discourse-configurational language where the configuration of constituents expresses information structural roles as opposed to grammatical ones (cf. [É. Kiss 1995](#)) – and various aspects of it has garnered significant attention in both theoretical and psycholinguistic research.

¹ Verbal prefixes in Hungarian can modify the meaning of the verb, or act as aspect markers. *Meg-* in this case has a perfective aspect meaning, indicating that the action has been completed.

² Linguistically focused elements in the examples will be set in capitals for highlight throughout the paper.

One of the key issues around Hungarian focus is whether it is inherently exhaustive, or if the exhaustivity is a pragmatic inference. In other words, does the PVF construction itself entail that the focused element is the only one for which the predicate holds true, or can it be interpreted in a contrastive way, where the focused element stands in opposition to other alternatives without necessarily excluding them? For example, in 2 above, does this sentence implies that Anna ate only the banana *and nothing else* – an exclusive reading, or could it mean that Anna ate the banana *and not something else* (or even, that she ate the banana but maybe *she also ate something else*) – a contrastive reading? We are interested in the ‘default’ interpretation of focus constructions such as these, and whether Hungarian speakers tend to understand them exhaustively or contrastively, demonstrated by the two possible readings given in 2.

We should therefore first try to define what we mean by *exhaustive focus* and *contrastive focus*. First, both types require the pre-verbal position, but they differ in their semantic/pragmatic implications. In É.Kiss (1998)’s influential paper, both would fall under the term of *identificational focus*, which she defines as a focus that identifies the referent of the focused element from a set of alternatives. It takes scope, binds a variable, and interacts with quantification. É. Kiss treats contrastive focus as a subtype of identificational focus, where she describes it as [+exhaustive], and often [+contrastive]. However, we would argue that it does not always entail exhaustivity. In Krifka (2008:259)’s work on Information Structure (IS), both exhaustive and contrastive focus is mentioned as subtypes of focus, and while exhaustive is interpreted as the only alternative that makes the proposition true narrowing the set of alternatives down to one (and thus enforcing exclusivity), contrastive focus is defined as a focus used for truly contrastive utterances and presupposes a competing alternative in the common ground, highlighting the contrast. Krifka’s distinction aligns with his broader distinction between semantic (truth-conditional impact) and pragmatic (discourse management) uses of focus.

In short, exhaustive focus implies that the focused element is the sole entity for which the predicate holds true; i.e. that there are no other things Anna ate. On the other hand, contrastive focus would imply that the focused element stands in opposition to other possible alternatives, drawing attention to its uniqueness, unlikelihood, or contradicting nature given the context, e.g., ‘Anna ate the banana [and not the apple].’, but it does not necessarily exclude alternatives, i.e., that there could be other things that Anna ate, e.g., ‘Anna ate the banana [and/but also the apple].’

- (3) Anna **csak** a BANÁNT ette meg.
 Anna[NOM] only the banana-[ACC] eat-3PST-SG.DEF VPX
 ‘Anna ate (up) only the BANANA.’

- (4) Anna **többek között** a BANÁNT ette meg.
 Anna[NOM] others amongst the banana-[ACC] eat-3PST-SG.DEF VPX
 ‘Anna ate (up) the BANANA, amongst other things.’

We can, in fact, force an exhaustive reading by adding the exhaustive focus marker *csak* ‘only’ to the sentence (ex. 3)³. Furthermore, we can also cancel exhaustivity within

³The scope of *csak* ‘only’ only extends to the subsequent element, it can only refer to the object and not the following VP.

the same clause with the help of *többek között* ‘amongst others’ (ex. 4)⁴, a phrase with an inherently non-exhaustive/inclusive meaning (see also Balogh 2013; Wedgwood 2006:14).

Now, one intuitive reading of 2 would be that ‘Anna ate the banana [and nothing else].’ In this sense, this usual PVF sentence seems to be equal to 3, where the exhaustive interpretation is explicitly expressed with the operator *csak* ‘only’. The question of differentiating the exhaustivity entailed by the pre-verbal focus structure vs the exhaustivity operator has been studied in detail by Balogh (2007). However, we also notice that if the utterance 2 is amended with certain phrases, e.g. *többek között* ‘amongst others’ as in 4, it is not only perfectly grammatical, it becomes inclusive, and also expresses a contrastive sense due the use of focus.

In Hungarian linguistic research, the role of semantic exhaustivity in focus constructions has been debated for more than forty years. The central question is whether exhaustivity arises from the syntax–semantics interface or from pragmatic and contextual factors. Earlier accounts within the generative framework assumed that the Hungarian pre-verbal focus structure necessarily entails semantic exhaustiveness, claiming that exhaustivity is an inherent property of the focus structure itself (É.Kiss 1998, 2007; Horvath 2008; Kene-sei 2006; Szabolcsi 1981) (the received view up until the 2000s). More recently, this view has since been challenged and shown to be untrue by both theoretical and psycholinguistic work – supported by experimental data – and it has been proposed that the exhaustive interpretation emerges from discourse-pragmatics, or conversational implicature (Balogh 2013; Káldi and Babarczy 2016; Onea 2009; Wedgwood et al. 2006). In introducing their argument against the claims of Szabolcsi (1981), É.Kiss (1998, 2007), and Horvath (2008) that ‘an immediately pre-verbal focus in Hungarian is semantically exhaustive’ or ‘is interpreted exhaustively, i.e. as if it were in the scope of an exclusive like “only”’. In their first example⁵ – analogous to our example 2 – Onea & Beaver (2009:1) suggested that the implicit reading of a sentence of this type of construction should be exhaustive, i.e. ‘Anna ate the banana [and nothing else].’ but their analysis suggests that while such sentences often invite an exhaustive reading, this interpretation is not structurally encoded but contextually inferred.

In 2 however, we feel that the underlying implication is not necessarily that Anna ate *nothing* else, but that she did not eat *something* else. Therefore, in our view, the default reading might as well be contrastive.

To test this, we designed an experiment where participants read typical PVF sentences and are forced to choose between two contexts that offer either an A) exhaustive interpretation, and B) a contrastive interpretation (and where both are plausible). We used a visual-world paradigm, where participants complete a sentence–picture matching/verification task after presented with a focused sentence in a self-paced reading manner, where they have to decide which image/context/situation best describes the sentence. Their choices...

To sum up, while previous work has asked whether an immediately pre-verbal focus in Hungarian is semantically exhaustive or not, our question is slightly different: What is the default interpretation of a pre-verbal focus construction? Is it inherently exhaustive, or is it more naturally contrastive?

⁴Similarly, only the object is in the scope of *többek között* ‘amongst others’, it is not possible to interpret this sentence as if the verb is an alternative to other actions one can do to a banana.

⁵*Péter MARIT csókolta meg.* ‘Peter kissed MARY.’

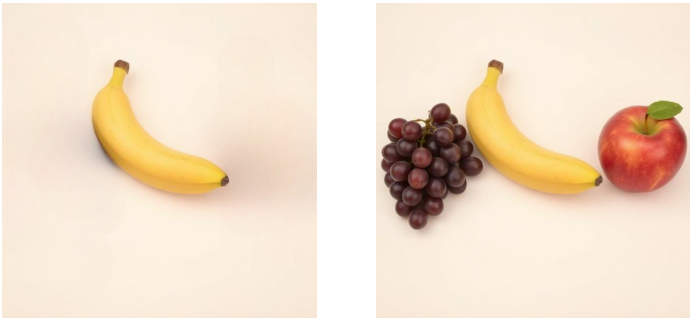


Figure 1. Example stimuli...

1.1. Background

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2. Methods

In this study, we are investigating focus in Hungarian, specifically how Hungarian speakers distinguish between exhaustive focus and contrastive focus – if they do. So... we designed an experiment

2.1. Participants

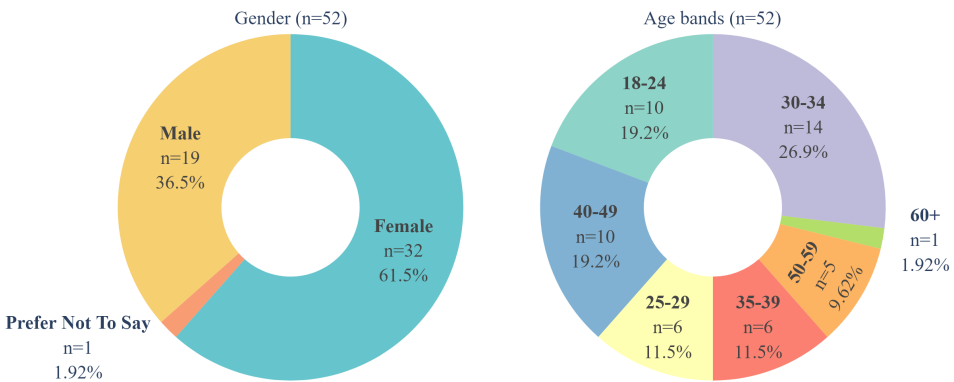


Figure 2. Demographic information of the participants in the experiment; gender (left) and age distributions (right).

We recruited 55 native speakers of Hungarian via the Prolific platform (www.prolific.com), where participants were pre-screened for Hungarian as their first language, and received a monetary compensation for their time (around £7.50/h). After rejecting 3 participants (due to overly long submission time, failing attention checks, or missing ET data),

the final sample consisted of 52 individuals (32 female; mean age = 35). On average it took people just under 23 minutes to complete the experiment, although with a high individual variation (very lengthy submission times were rejected).

2.2. Materials

- (5) a. Ami az innivalót illeti, Eszter csak kávéét ivott.
which[REL] the drink-ACC regard-3SG Eszter only coffee-ACC drink-PST-3SG
‘Regarding drinks, Eszter only drank coffee.’
- b. Ami az innivalót illeti, Eszter ma kávéét ivott.
which[REL] the drink-ACC regard-3SG Eszter today coffee-ACC drink-PST-3SG
‘Regarding drinks, Eszter drank coffee today.’
- c. Ma reggel Eszter többek között kávéét ivott.
today morning Eszter[NOM] amongst others coffee-ACC drink-PST-3SG
‘This morning, Eszter drank, amongst other things, coffee.’



Figure 3. Example stimuli...

Articles were kept together with nouns when presented on the screen, so all 3 conditions had the same number of regions (7), and the focused element was always in region 6 (r6). The stimuli were created by modifying a set of 18 base sentences, which were all simple transitive sentences with a subject, verb, and direct object. The three conditions were created by modifying the base sentences with the addition of the focus marker *csak* ‘only’ for the exhaustive condition, and *többek között* ‘amongst others’ for the contrastive condition.....

The stimuli consisted of 54 experimental sentences (18 items × 3 conditions) and 18 corresponding image pairs with 54 filler sentences in a randomized order, but making sure that the same items never followed each other. The full stimuli list is available online ??.

To avoid overly long experimental sessions, we divided the experiment into three equal parts and each participant could complete one, two, or all three parts as separate events; meaning that we had three participant groups that could overlap. This resulted in 72 valid submissions (25 for group a, 23 for group b, and 24 for group c), thus every stimuli was

interacted with at least 23 times. The three parts were split in a way that for each group each item only appeared in one condition, and each session contained four attention check questions intermixed and evenly distributed across the trials.

The three conditions were: (i) simple pre-verbal focus (PVF), (ii) exhaustive focus with *csak* ‘only’, and (iii) contrastive focus with *többek között* ‘amongst others’. Each sentence was paired with two images: one depicting a single-object interpretation (type A) and one depicting a multiple-object interpretation (type B).

EXAMPLE

2.3. Design

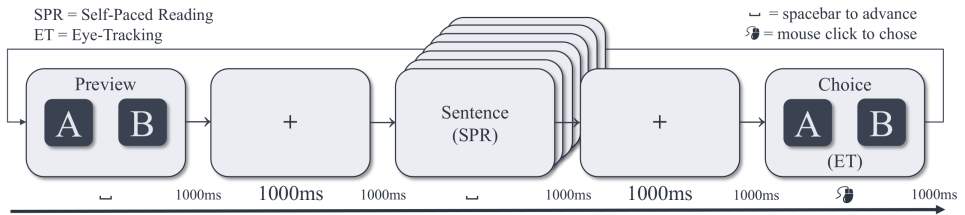


Figure 4. Design

After a practice session consisting of three trials, participants saw a preview of the image pair to get familiar with the objects/context, on the left and right half of the screen (20% of the screen's width and 40% of the screen's height) in a canvas 50% larger than the images. After pressing the spacebar they were presented with the sentence word by word (region by region), in the middle of the screen in a self-paced reading (SPR) manner participants had to use the spacebar to advance through the sentences. We recorded reaction times (RT) throughout the reading, and the focused elements appeared in region six (r6) in every condition. After reading through the sentence, the image pair re-appeared and participants were asked to select the image that best matched the meaning of the sentence they just read. The choice had to be made with a mouse-click anywhere on the corresponding canvas. Reaction times and eye movements were recorded during this picture verification phase.

Eye tracking what

Images type A and B were counterbalanced throughout the trials.

2.4. Procedure

The experiment was hosted on and presented using PCIBex (Schwarz & Zehr 2021), the code and resources are available online ??

Participants completed the experiment on their own devices (PC only with a webcam and physical keyboard & mouse) and own their own time, but with Prolific helping us control for too quick or too slow submissions resulting in rejection/timeout.

Participants were instructed to be in a well-lit environment, sit at a comfortable distance from their screen and webcam, and minimize head movements during the experiment, and were informed about the exact procedure.

Participants also had to calibrate the eye-tracker before starting the experiment. After calibration, practice trials were provided to familiarize participants with the self-paced reading and picture verification tasks.

Eye-tracking data were collected using the WebGazer.js library built in PCLib, which uses the participant's webcam to track eye movements. Before starting the experiment, participants underwent a calibration procedure to ensure accurate eye-tracking data collection.

2.5. Data Analysis

Reading times and dwell times were analyzed using linear mixed-effects models with FocusCondition, ChoiceType, and their interaction as fixed effects, and Participant and Item as random effects. Post-hoc comparisons were conducted using Tukey HSD tests. Trials with incorrect responses or extreme RTs (>2.5 SD from the mean) were excluded.

3. Results

4. Discussion

5. Conclusion

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Unnumbered section

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Data availability statement. A statement about how to access data, code and other materials allowing users to understand, verify and replicate findings – e.g. Replication data and code can be found in Harvard Dataverse: <https://doi.org/link>.

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